

CSE431: Mobile Programming Course Project

Presented to Dr. Haitham Azmi

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1 Github Repo

<https://github.com/omar4a/MP.git>

2 YouTube Demo Video

<https://youtu.be/any7uS02T5A>

3 Introduction

Mind Metric is a Flutter app for real-time EEG monitoring and cognitive wellness. It connects to a Python bridge that streams data to Firebase Realtime Database. The app shows live status, band powers (delta, theta, alpha, beta, gamma), mental state, and supports calibration, scheduling, alerts, and analytics. It also supports optional text-to-speech (TTS) and speech commands for accessibility.

4 Survey of Similar Apps

We compared two popular EEG software suites: Emotiv and Neurable.

Emotiv Software Suite (EmotivPRO / EmotivBCI)

Advantages

- Strong ecosystem and research focus.
- Supports multiple devices and data export.
- Detailed signal tools and dashboards.

Disadvantages

- Cost and subscription is high.
- Interface can be complex for new users.
- Some features are locked behind paid tiers.

Neurable Software Suite

Advantages

- Focus on hands-free control and productivity.
- Tight integration with Neurable hardware (headphones).
- Simple focus and state metrics for users.

Disadvantages

- Limited hardware availability.
- Less access to raw EEG data compared to research tools.
- Smaller ecosystem and fewer advanced analysis tools.

5 UI Design (Screens and Placeholders)

Splash Screen

Short branding screen with logo and loading indicator.

Login Screen

User email + password login and password recovery.

Register Screen

User registration with name, email, password.

Password Recovery Screen

Password reset by email.

Dashboard Screen

Shows system status (streaming or offline), band powers, mental state, and Start/Stop Session buttons.

EEG Monitor Screen

Shows per-channel EEG sensor quality.

Control Panel Screen

Start/Stop/Recalibrate and scheduling UI.

Calibration Screen

Per-session calibration steps with progress bar.

Analytics & History Screen

Charts, trend analysis, and export options.

Alerts Screen

Persistent history of notifications.

Settings Screen

User profile display, notification toggle, speech toggles, EEG device dropdown, sign out.

Figure 1: Splash Screen

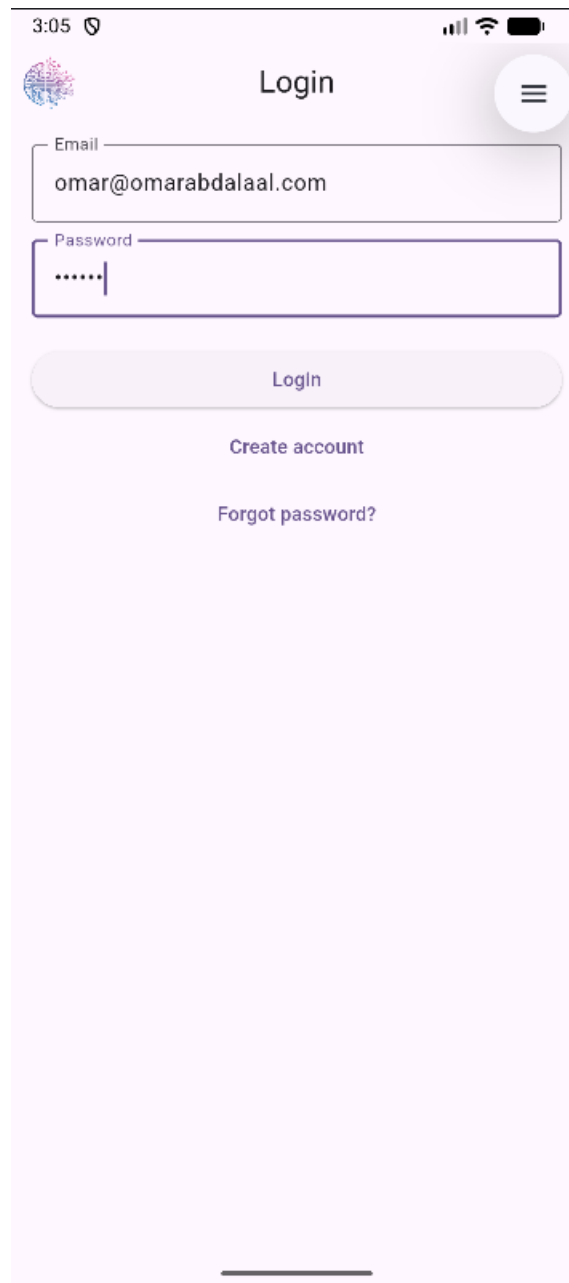


Figure 2:

Figure 3:

Figure 4:

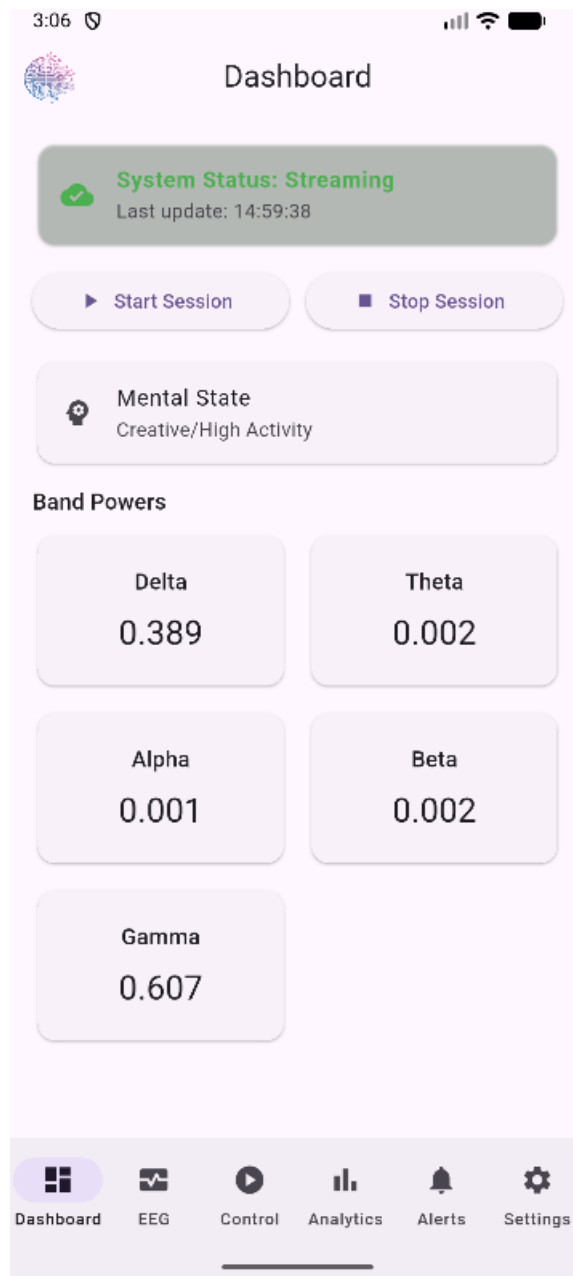


Figure 5:

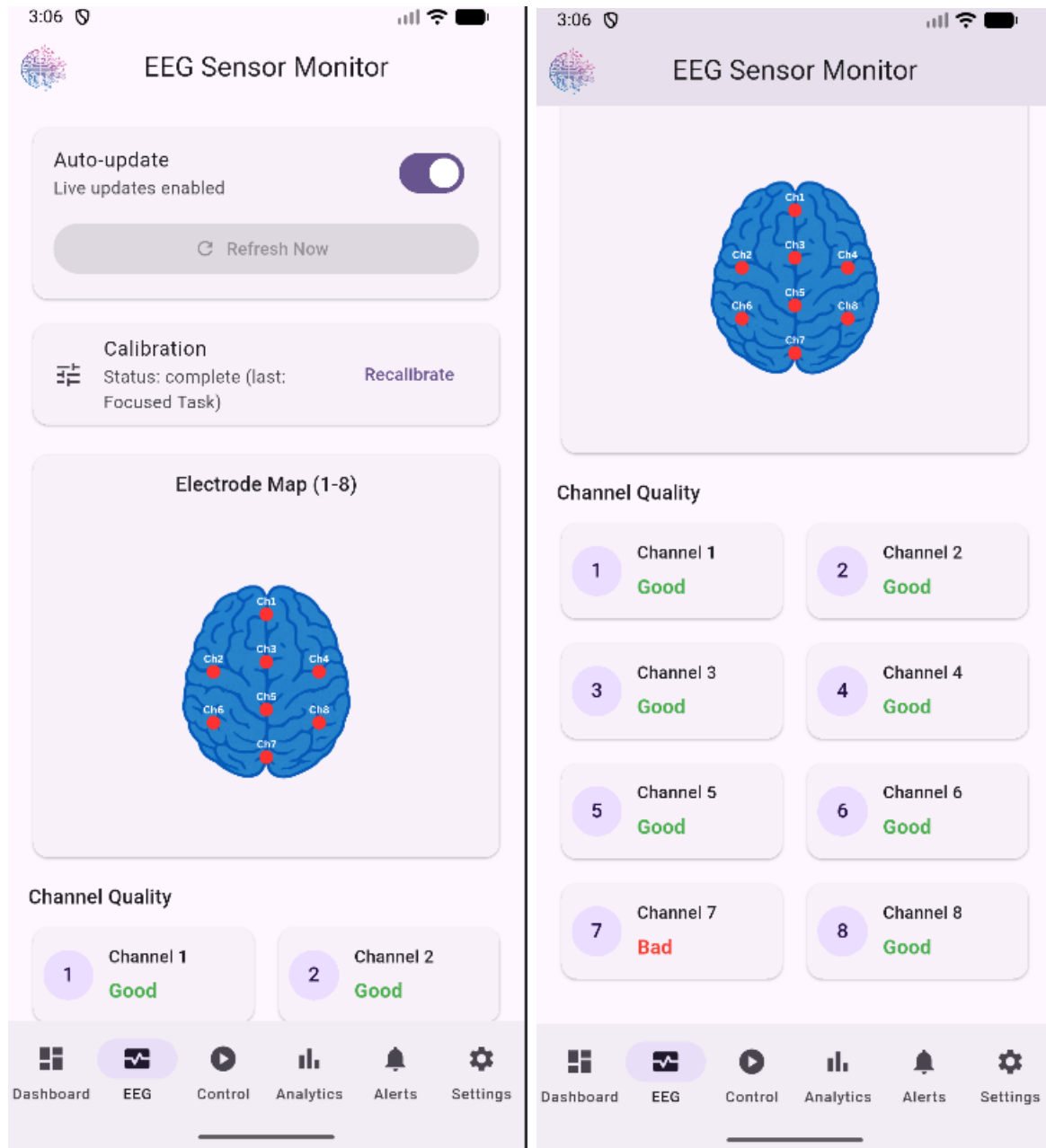


Figure 6:

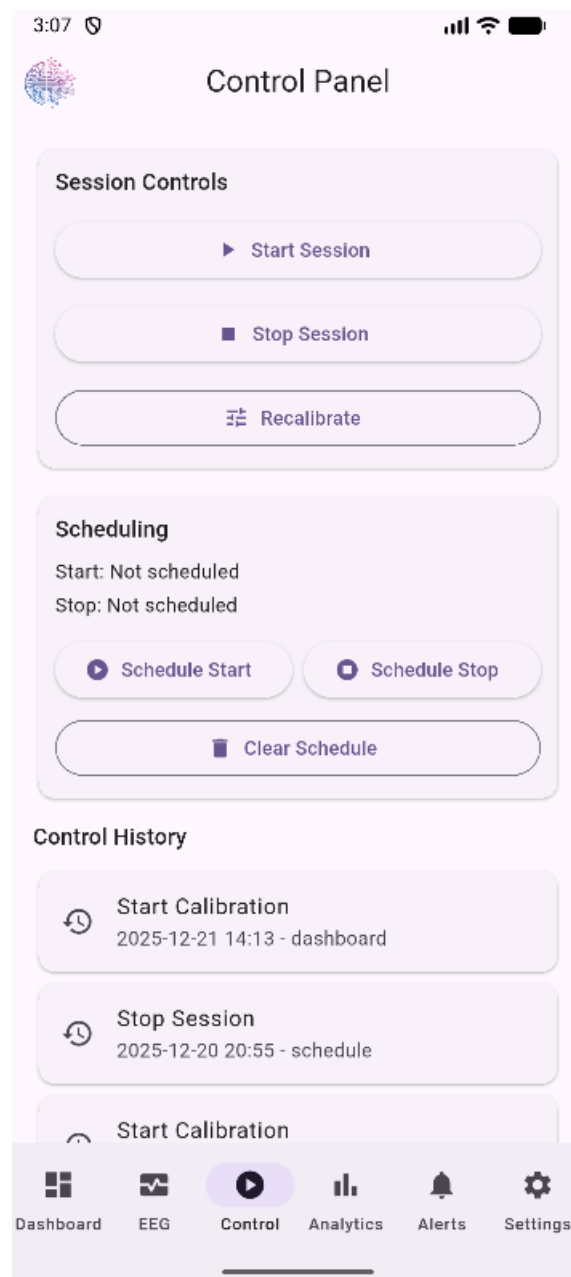


Figure 7:

Figure 8:

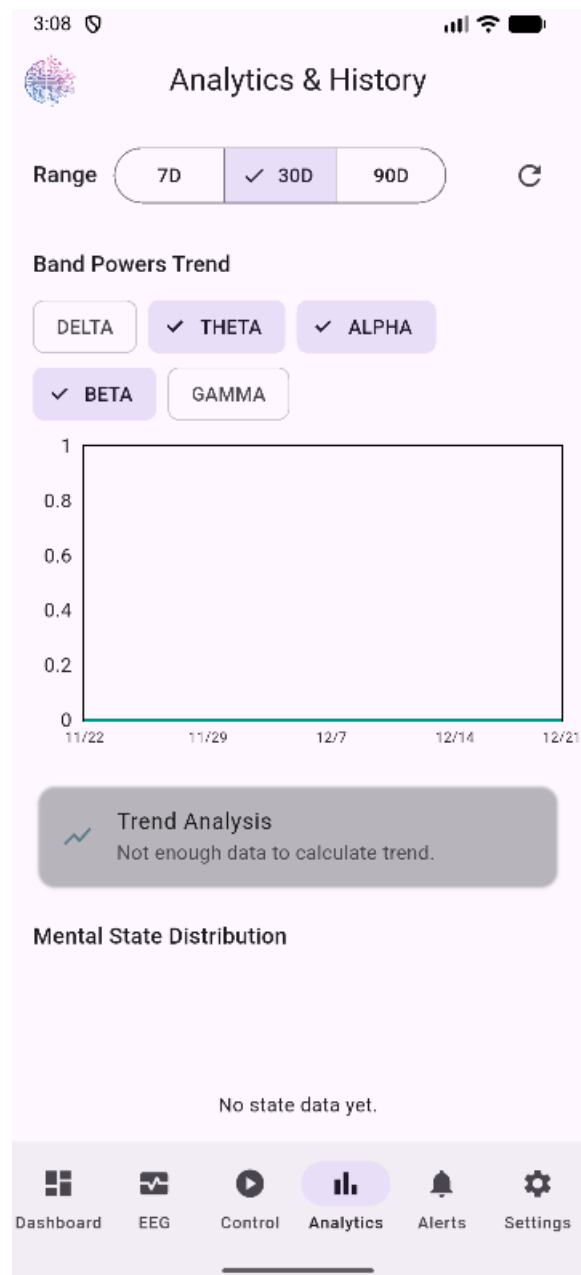


Figure 9:

Figure 10:

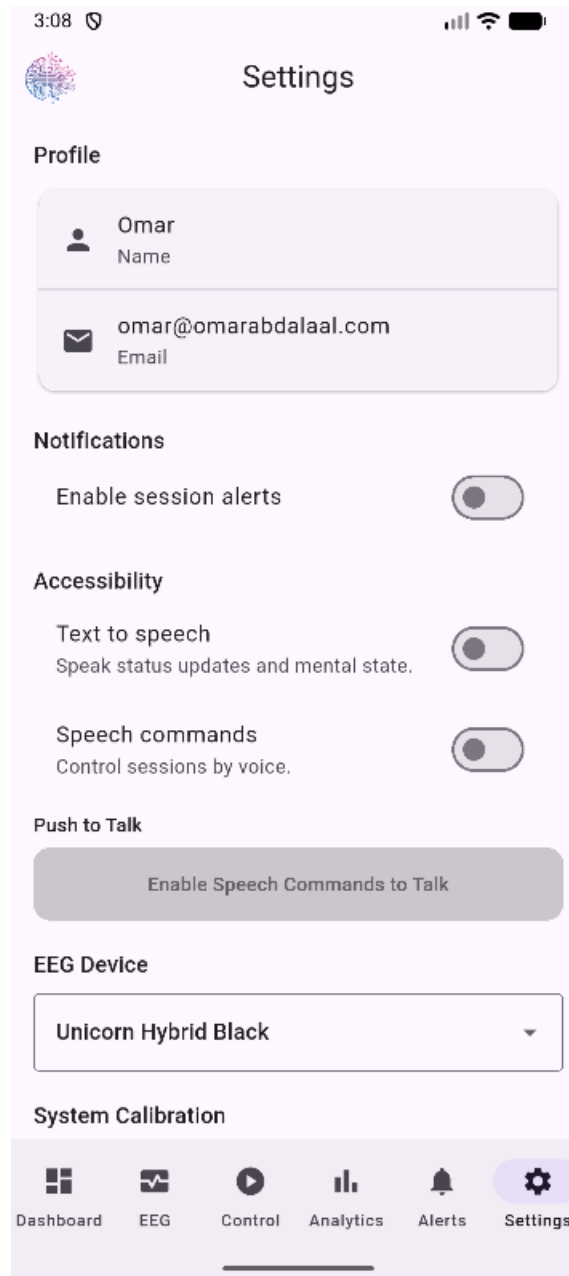


Figure 11:

6 Code Description and Navigation Scenarios

Code Description

- **Flutter App** in `lib/` with screens, services, and providers.
- **Python EEG Bridge** in `lib/bridge.py` streams EEG to Firebase RTDB.
- **Realtime Database** for live data, commands, schedules, and history.
- **SQLite** for local history logs (alerts, sensor activity, schedule logs).
- **Provider** for local state management and alerts list.
- **Tests** include unit, widget, and integration tests.

Navigation Scenarios

Scenario 1: Start Session

- User logs in.
- Dashboard shows status.
- User taps *Start Session*.
- Calibration screen opens and runs steps.
- After calibration, streaming starts.

Scenario 2: Recalibrate While Streaming

- User is streaming.
- User taps *Recalibrate* in Control Panel.
- Calibration starts; streaming continues for calibration.

Scenario 3: Scheduled Start/Stop

- User opens Control Panel and schedules a time.
- At the time, a push notification is sent.
- App starts calibration and then streaming.
- Stop time triggers push notification and stop.

7 Test Plan and Scoreboard

Test Plan

- **Unit Tests:** Speech command parsing, local SQLite retention.
- **Widget Tests:** Dashboard, Control Panel, Calibration UI.
- **Integration Tests:** Register/login flow, start session flow, schedule flow.

Scoreboard of Testcases

Automated Test Script: `scripts/run_tests.ps1` **Logs:** `test_logs/test_run_*.log` and `test_logs/logcat_*.txt`

8 Known Bugs

- No critical bugs known at this time.
- Voice features may be limited on emulators.
- Analytics will show no data until sessions generate history.

9 Version Control Activity Report (GitHub)

- Total commits: 5
- Authors: Omar
- Recent commits:
 - `f711624` final adjustments
 - `47834f6` adjustments + local DB & speech support
 - `bc9dc02` calibration, FCM, schedule
 - `0c75306` bridge + sensor status functional
 - `f6f1e6d` Milestone 1

	IXIII	ID	Testcase	Type	Result	Evidence
	T01		Speech command parsing	Unit	Pass	<code>flutter test</code>
	T02		SQLite retention purge	Unit	Pass	<code>flutter test</code>
	T03		Dashboard UI renders	Widget	Pass	<code>flutter test</code>
	T04		Control Panel schedule UI	Widget	Pass	<code>flutter test</code>
	T05		Calibration first step UI	Widget	Pass	<code>flutter test</code>
T06			Register + Start Session	Integration	Pass	<code>scripts/run_tests.ps1</code>
T07			Schedule start/stop flow	Integration	Pass	<code>scripts/run_tests.ps1</code>